

Zepto Inventory Analysis

An end-to-end SQL analytics project exploring product pricing, discount strategies, inventory weight distribution, revenue estimation, and stock availability using the Zepto Inventory Dataset from Kaggle.

Metric	Value
Dataset Source	Kaggle — Zepto Inventory Dataset
Total Records (after cleaning)	~3,200+ SKUs
Product Categories	14
Database	PostgreSQL
Analysis Queries	8 Business Questions

1. Project Overview

This project analyzes the Zepto grocery delivery platform's inventory data to extract actionable business insights. Zepto operates as a quick-commerce platform, and understanding its product mix, pricing, discounting patterns, and stock availability is critical for operational and marketing decisions.

The key business questions answered through this analysis are:

- Which products offer the best value through highest discount percentages?
- Which high-MRP products are currently out of stock — representing lost revenue?
- What is the estimated revenue potential for each product category?
- Which premium-priced products carry minimal discounts?
- Which categories offer the highest average discounts to customers?
- Which products offer the best price-per-gram value for consumers?
- How are products distributed across weight categories (Low / Medium / Bulk)?
- What is the total inventory weight burden per category?

2. Dataset Summary

The dataset is sourced from Kaggle (Zepto Inventory Dataset by palvinder2006) and contains real product-level inventory data from Zepto's quick-commerce grocery platform.

Attribute	Details
Source	Kaggle — zepto_v2.csv
Total Rows	~3,200+ product SKUs
Total Columns	10
Missing Data	Checked via NULL queries — none critical

Key Columns

- **sku_id** — Unique product identifier (SERIAL PRIMARY KEY)
- **category** — Product category (e.g., Fruits & Vegetables, Beverages)
- **name** — Full product name
- **mrp** — Maximum Retail Price (in paise → converted to rupees)
- **discountPercent** — Discount percentage offered
- **discountedSellingPrice** — Final selling price after discount
- **weightInGms** — Product weight in grams
- **availableQuantity** — Units available in inventory
- **outOfStock** — Boolean flag for stock availability
- **quantity** — Pack quantity

3. Data Exploration & Cleaning using PostgreSQL

All data exploration and cleaning was performed directly inside PostgreSQL using pgAdmin. The following steps were executed before business analysis:

Data Exploration

- **Row Count:** Confirmed total number of records using `SELECT COUNT(*) FROM zepto;`
- **Sample Inspection:** Viewed the first 10 rows using `SELECT * FROM zepto LIMIT 10;` to understand structure and content.
- **NULL Check:** Queried all columns for NULL values — no critical missing data was found.
- **Category Discovery:** Used `SELECT DISTINCT category` to list all 14 unique product categories.
- **Stock Status:** Grouped by `outOfStock` to count in-stock vs out-of-stock products.
- **Duplicate SKUs:** Identified product names appearing multiple times (same product in different sizes/packs) using `HAVING COUNT(sku_id) > 1`.

Data Cleaning

- **Removed Zero-Price Products:** Found and deleted records where `mrp = 0`, as these were invalid entries that would skew price analysis.
- **Converted Paise to Rupees:** The original dataset stored prices in paise (1/100th of a rupee). Applied an `UPDATE` to divide both `mrp` and `discountedSellingPrice` by 100 to convert to standard Indian Rupee (INR) values.

4. Data Analysis using SQL (PostgreSQL)

Eight structured SQL queries were written to extract business insights from the cleaned dataset. Each query targets a specific business question with results shown below.

Q1. Top 10 Best-Value Products by Discount Percentage

Identified the 10 products with the highest discount percentages to understand which items Zepto is promoting most aggressively. Dukes Waffly range leads with 51% discount, followed by multiple products at 50% off.

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	45.00	51.00
2	Dukes Waffy Chocolate Wafers	45.00	51.00
3	Dukes Waffy Strawberry Wafers	45.00	51.00
4	RRO Mozzarella Block Cheese	295.00	50.00
5	Ceres Foods Laal Maas Instant Liquid Mas...	220.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00
7	RRO Mozzarella Pizza Cheese	275.00	50.00
8	Ceres Foods Fish Mustard Instant Liquid M...	220.00	50.00
9	Dukes Waffy Strawberry Roll	150.00	50.00
10	Ceres Foods Nalli Nihari Instant Liquid Mas...	220.00	50.00

Q2. High-MRP Products That Are Out of Stock

Found premium products (MRP > Rs.300) that are currently out of stock — these represent direct lost revenue opportunities. Patanjali's Cow's Ghee (Rs.565) tops the list, followed by MamyPoko Diapers (Rs.399) and Aashirvaad Atta (Rs.315).

	name character varying (150)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	565.00
2	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
3	Aashirvaad Atta With Mutigrains	315.00
4	Everest Kashmiri Lal Chilli Powder	310.00

Q3. Estimated Revenue by Product Category

Calculated total estimated revenue per category using discounted selling price multiplied by available quantity. Personal Care leads with the highest revenue potential, followed by Home & Cleaning and Ice Cream & Desserts. Fruits & Vegetables and Meats show the lowest revenue numbers due to lower price

points.

	category character varying (120) 🔒	total_revenue numeric 🔒
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00
6	Health & Hygiene	64180.00
7	Home & Cleaning	122661.00
8	Ice Cream & Desserts	224385.00
9	Chocolates & Candies	224385.00
10	Packaged Food	224385.00
11		
Total rows: 14		Query complete 00:00:00.11

Q4. Products with MRP > Rs.500 and Discount < 10%

Identified premium products priced above Rs.500 that carry minimal discounts (less than 10%). These are typically high-margin items where Zepto does not need aggressive discounting — primarily edible oils (Dhara, Saffola, Fortune) and Surf Excel detergents.

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dhara Kachi Ghani Mustard Oil Jar	1250.00	8.00
2	Saffola Gold (Jar)	1240.00	0.00
3	Dhara Filtered Groundnut Oil (Jar)	1050.00	1.00
4	Fortune Rice Bran Health Oil (Jar)	1050.00	1.00
5	Dhara Filtered Groundnut Oil (Jar)	1050.00	0.00
6	Fortune Soyabean Oil	1005.00	0.00
7	Fortune Sunlite Refined Sunflower (Jar)	925.00	0.00
8	Surf Excel Matic Powder Front Load	810.00	7.00
9	Surf Excel Matic Top Load	720.00	9.00
10	Pedigree Puppy Dry Dog Food Food Chicken & Milk	690.00	6.00
11	Pedigree Puppy Dry Dog Food Food Chicken & Milk	690.00	6.00
Total rows: 39		Query complete 00:00:00.094	CRLF Ln 102

Q5. Top 5 Categories by Average Discount Percentage

Ranked product categories by their average discount offered to customers. Fruits & Vegetables lead with an average 15.46% discount, suggesting competitive pricing strategy in perishables. Meats, Fish & Eggs follow at 11.03%, while Ice Cream, Chocolates, and Packaged Food all offer around 8.32% average discount.

	category character varying (120)	avg_discount numeric
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

Q6. Price Per Gram for Products Above 100g (Best Value Sort)

Calculated price-per-gram for all products weighing 100g or more, then sorted ascending to surface the best value-for-money items. Products like Onion and basic staples rank at the top as the most economical per-gram options, useful for price-sensitive consumer targeting.

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)
1	Onion	3000	57.00
2	Onion	1000	21.00
3	Aashirvaad Iodised Salt	1000	19.00
4	Vicks Cough Drops Menthol	1160	20.00
5	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	28.00
6	Tata Salt	1000	24.00
7	Baby Potato	500	16.00
8	Potato	3000	84.00
9	Potato	1000	29.00
10	Shubh kart - Tejas Twisted Cotton Wicks 1000n	1000	28.00
Total rows: 1395		Query complete 00:00:00.110	CRLF Ln 116, Col 1

Q7. Product Weight Categorization (Low / Medium / Bulk)

Grouped all products into three weight tiers using a CASE statement: Low (under 1,000g), Medium (1,000g–5,000g), and Bulk (5,000g+). The vast majority of products fall into the Low category, reflecting Zepto's focus on small daily-use pack sizes for quick-commerce delivery.

	name character varying (150)	weightInGms integer	weight_cat text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low
7	Sapota	348	Low
8	Britannia Treart Creme Wafers Chocolate	60	Low
9	Popular Essentials Dhania (Coriander Seeds)	200	Low
10	Popular Essentials Californian Almond	250	Low
Total rows: 1782 Query complete 00:00:00.255 CRLF Ln 123, Col 1			

Q8. Total Inventory Weight Per Category

Calculated the total physical inventory weight per category by multiplying weightInGms by availableQuantity. Chocolates & Candies and Ice Cream & Desserts carry the heaviest inventory load, while Meats, Fish & Eggs and Biscuits are the lightest — indicating how warehouse space is distributed across categories.

	category character varying (120) 	total_weight bigint 
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187
9	Home & Cleaning	373161
10	Ice Cream & Desserts	490797
11	Others	490797
Total rows: 14		Query complete 00:00:00.12

5. Business Recommendations

Restock High-MRP Out-of-Stock Products

- Premium items like Patanjali Ghee (Rs.565) and MamyPoko Diapers (Rs.399) are out of stock — restoring inventory here recovers high-margin revenue immediately.

Reassess Heavy Discounting on Waffy & Cheese Products

- Products discounted at 50–51% (Dukes Waffy range, RRO Mozzarella) may be eroding margins. Consider capping discounts at 35–40% or running time-limited offers instead.

Target Fruits & Vegetables for Loyalty Campaigns

- This category offers the highest average discount (15.46%) and is a daily-purchase driver. Bundle deals or subscription offers here can significantly boost repeat orders.

Promote Premium Oils as Full-Price Anchors

- High-MRP oils (Dhara, Saffola, Fortune) priced above Rs.1,000 carry near-zero discounts and still sell — use these as margin anchors while discounting in adjacent categories.

Optimize Warehouse Space Using Weight Data

- Chocolates & Ice Cream carry the heaviest inventory weight. Reviewing reorder quantities for these vs. fast-moving light categories like Meats can improve storage efficiency.